

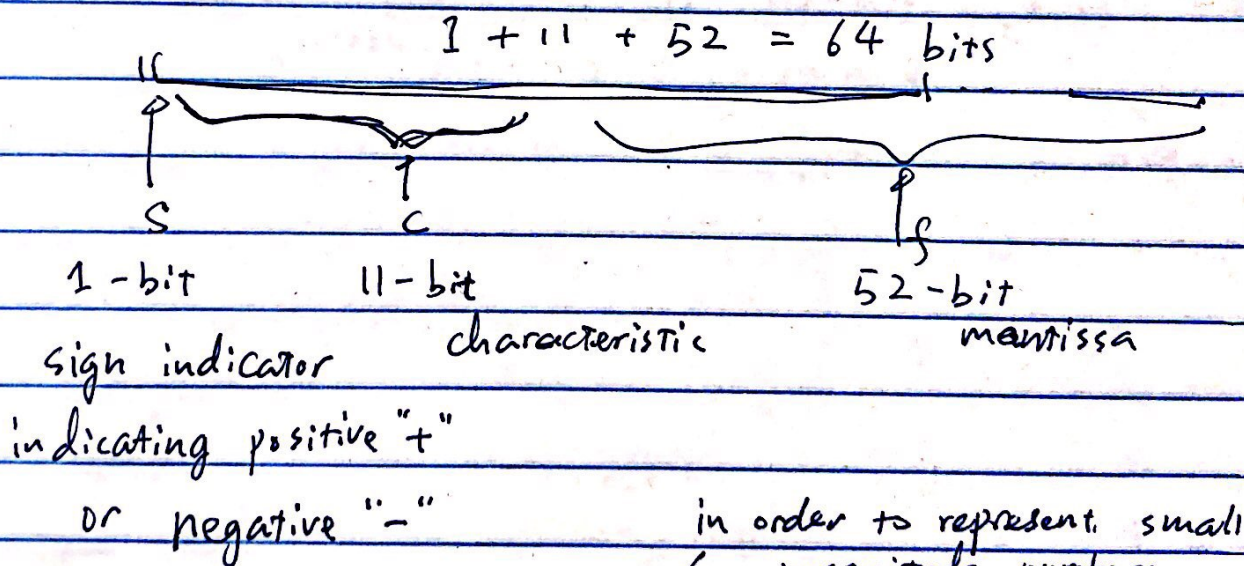
Review.

1.2. Computer Arithmetic

computer is using finite digit numbers.

↳ comparing with real number
round-off error

64-bit (binary digit) representation
by IEEE standard is



in order to represent small magnitude numbers.

$$(-1)^s 2^{C-1023} (1+f)$$

this is how binary digit representation in computer
convert into a decimal digit number

(check Illustration on Pg 15)

Largest number in a 64-bit machine:

$$2^{1023} \cdot (2 - 2^{-52}) \approx 0.18 \times 10^{309}$$

here, sign indicator $S = 0$

characteristic $C = 2048$ (2^{11})

mantissa ~~$f = 1 - 2^{-52}$~~

$$f = 1 - 2^{-52} (2^{-1} + 2^{-2} + 2^{-3} + \dots + 2^{-52})$$

any number larger than $2^{1023} \cdot (2 - 2^{-52})$ in machine would cause overflow, making computation stop.

Smallest number in a 64-bit machine: ~~two~~

$$2^{-1022} \cdot (1 + 0) \approx 0.22 \times 10^{-307}$$

here, sign indicator $S = 0$ (smallest in magnitude!)

characteristic $C = 1$

mantissa $f = 0$

any number smaller than 2^{-1022} in this machine would cause underflow, making number to 0.
(while it's actually not 0)